Source	$\max|\Re(\lambda)|$	$\max|\Im(\lambda)|$	$\rho(L)$	$\kappa_{\neq 0}(L)$
DienerDorbandSchnetterTiglio2007 [1]	8.579989e+03	0.000000e+00	8.579989e+03	4.731345e+14
Mattsson2012 [2]	8.579989e+03	0.000000e+00	8.579989e+03	4.731345e+14
Mattsson2014 [3]	1.219214e+05	7.262608e+00	1.219214e+05	4.646055e+14
Mattsson2017 [4]	1.253174e+07	0.000000e+00	1.253174e+07	1.969298e+16
MattssonAlmquistCarpenter2014Extended [5]	8.579989e+03	0.000000e+00	8.579989e+03	2.023431e+13
MattssonNordström2004 [6]	8.579989e+03	0.000000e+00	8.579989e+03	4.731345e+14
MattssonSvärdNordström2004 [7]	8.579989e+03	0.000000e+00	8.579989e+03	4.731345e+14
MattssonSvärdShoeybi2008 [8]	8.579989e+03	0.000000e+00	8.579989e+03	4.731345e+14
WilliamsDuru2024 [9]	8.875274e+06	1.064645e+02	8.875274e+06	1.269667e+16

Table 1: Laplacian spectral metrics for successful SBP sources (method=diagonal, N=31, order=4, p=2, R=1.0). Conditioning uses all eigenvalues with abs(lambda) != 0 exactly (zero tolerance = 0.0).

References

[1] Diener, Dorband, Schnetter, and Tiglio. Optimized high-order derivative and dissipation operators satisfying summation by parts, and applications in three-dimensional multi-block evolutions. *Journal of Scientific Computing*, 32(1):109–145, 2007.

[2] Mattsson. Summation by parts operators for finite difference approximations of second-derivatives with variable coefficients. *Journal of Scientific Computing*, 51:650–682, 2012.

[3] Mattsson. Diagonal-norm summation by parts operators for finite difference approximations of third and fourth derivatives. *Journal of Computational Physics*, 274:432–454, 2014.

[4] Mattsson. Diagonal-norm upwind sbp operators. *Journal of Computational Physics*, 335:283–310, 2017.

[5] Mattsson, Almquist, and Carpenter. Optimal diagonal-norm sbp operators. *Journal of Computational Physics*, 264:91–111, 2014.

[6] Mattsson and Nordström. Summation by parts operators for finite difference approximations of second derivatives. *Journal of Computational Physics*, 199:503–540, 2004.

[7] Mattsson, Svärd, and Nordström. Stable and accurate artificial dissipation. *Journal of Scientific Computing*, 21(1):57–79, 2004.

[8] Mattsson, Svärd, and Shoeybi. Stable and accurate schemes for the compressible navier-stokes equations. *Journal of Computational Physics*, 227:2293–2316, 2008.

[9] Williams and Duru. Full-spectrum dispersion relation preserving summation-by-parts operators. *SIAM Journal on Numerical Analysis*, 62(4):1565–1588, 2024.